

MALAGA AP, Spain

WMO: 084820

Lat: 36.666N

Lon: 4.482W

Elev: 16

StdP: 101.14

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.2	5.8	-4.8	2.5	11.5	-2.1	3.2	12.1	12.0	13.9	11.1	14.5	4.5	300	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	35.6	20.1	33.2	20.1	31.2	20.1	24.2	28.6	23.6	28.0	23.0	27.4	5.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	17.6	27.0	22.1	16.8	26.6	21.5	16.2	26.2	72.9	28.9	70.5	28.0	68.2	27.4	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	9.0	8.0	DB	1.8	39.6	1.3	1.6	0.9	40.8	0.1	41.7	-0.6	42.6	-1.6	43.8	(4)
			WB	0.2	25.1	1.5	1.0	-0.9	25.8	-1.8	26.4	-2.6	26.9	-3.7	27.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.8	12.6	13.1	14.7	16.7	19.6	23.4	25.9	26.5	23.6	20.0	15.8	13.4
	DBStd	5.45	2.36	2.35	2.32	2.34	2.54	2.42	2.42	2.22	2.40	2.36	2.75	2.23
	HDD10.0	11	5	3	1	0	0	0	0	0	0	0	1	2
	HDD18.3	763	178	146	114	60	14	0	0	1	11	86	154	
	CDD10.0	3229	86	90	148	201	299	403	494	511	409	310	173	106
	CDD18.3	940	0	1	3	11	55	153	236	252	159	62	8	0
	CDH23.3	7718	1	2	19	65	312	1162	2283	2514	1107	232	21	0
CDH26.7	2669	0	0	2	8	68	365	901	987	301	34	1	0	
Wind	WSAvg	3.6	4.3	4.1	3.8	3.6	3.5	3.5	3.4	3.3	3.1	3.0	3.8	4.0
Precipitation	PrecAvg	520	62	58	59	44	26	7	1	3	29	62	80	90
	PrecMax	1093	307	224	248	144	111	81	4	19	143	194	224	325
	PrecMin	286	3	0	0	0	0	0	0	0	0	0	6	1
	PrecStd	207	72	52	52	32	28	16	1	5	34	46	56	97
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	23.1	25.8	27.8	31.2	35.5	38.8	38.4	34.7	30.2	25.9	21.5
		MCWB	13.8	14.4	15.0	16.7	18.2	19.8	20.6	20.6	20.0	19.2	16.6	14.9
	2%	DB	19.5	20.8	22.9	25.1	28.6	32.6	35.9	35.2	31.7	27.1	23.2	20.0
		MCWB	13.3	13.7	14.5	15.6	17.2	19.3	20.1	20.3	19.8	18.8	16.3	14.4
	5%	DB	18.1	19.0	21.1	23.2	26.4	30.2	33.1	33.1	29.7	25.8	21.8	18.7
		MCWB	12.7	13.1	13.9	15.1	16.9	18.9	19.8	20.7	20.1	19.0	16.0	13.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	17.0	17.7	19.5	21.9	24.8	28.7	30.9	31.0	28.1	24.2	20.2	17.5
		MCWB	12.4	12.6	13.6	14.7	16.6	18.8	20.4	20.9	20.4	18.5	15.2	13.3
	0.4%	WB	15.6	16.0	17.0	18.5	20.2	22.9	25.0	25.2	24.1	21.5	19.2	16.7
		MCDB	19.1	20.7	21.7	24.0	25.7	28.1	29.1	29.8	28.6	26.0	23.0	19.4
	2%	WB	14.5	14.9	15.8	17.3	19.2	21.9	23.9	24.5	23.2	20.8	17.9	15.6
		MCDB	17.8	18.8	20.4	22.1	24.5	27.1	28.3	28.9	27.3	24.8	21.4	18.3
Mean Daily Temperature Range	5% DB	WB	13.7	14.0	15.1	16.4	18.5	21.1	23.2	24.0	22.5	20.1	17.0	14.8
		MCDB	16.9	17.4	19.3	21.1	23.6	26.4	27.6	28.4	26.5	23.8	20.2	17.5
	5% WB	WB	12.9	13.3	14.4	15.7	17.8	20.5	22.6	23.4	21.8	19.5	16.2	14.0
		MCWBR	6.3	6.6	6.2	6.0	5.5	5.5	5.3	5.3	5.3	5.0	5.2	5.6
Clear-Sky Solar Irradiance	taub	0.321	0.347	0.385	0.390	0.407	0.429	0.426	0.435	0.413	0.383	0.345	0.326	
	taud	2.528	2.467	2.355	2.359	2.341	2.292	2.295	2.274	2.364	2.455	2.522	2.536	
	Ebn at Noon	874	888	880	892	878	855	855	839	840	836	837	843	
	Edh at Noon	79	95	116	121	126	132	131	131	114	95	80	74	
All-Sky Solar Radiation	RadAvg	2.71	3.51	4.71	5.90	6.99	7.89	7.87	6.95	5.42	3.92	2.85	2.35	
	RadStd	0.23	0.40	0.50	0.41	0.47	0.23	0.18	0.21	0.30	0.32	0.24	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.44	N/A	N/A	N/A	+0.36	N/A	N/A	N/A	+169	+125
Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+63

Nomenclature: See separate page